

Homework 1

Handed Out: 01/15/2026

Due: 01/29/2026

1. (10 points) Let C be the solution set of a quadratic inequality:

$$C = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c \leq 0\},$$

where $\mathbf{b} \in \mathbb{R}^d, c \in \mathbb{R}, \mathbf{A} \in \mathbb{R}^{d \times d}$ is a symmetric matrix. Show that C is convex if $\mathbf{A} \succ 0$.

2. (10 points) Show that

$$f(x) = \frac{1}{2} \|x\|_2^2, x \in \mathbb{R}^d,$$

is 1-strongly convex w.r.t. $\|\cdot\|_2$.

3. (20 points) Recall the definition of p -norms as

$$\|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$$

for $p \geq 1$.

(1) (10 points) Show that 2-norms are always convex.

(2) (10 points) Show that p -norms are always convex. [Hint: Apply Minkowski inequality]

4. (20 points) Recall the Online Gradient Descent algorithm in lecture 3. We know that setting a constant learning rate $\eta^* = \frac{D}{L\sqrt{T}}$ gives us regret $DL\sqrt{T}$. We can achieve similar regret with a dynamic learning rate.

(1) (10 points) Prove that $\sum_{t=1}^T \frac{1}{\sqrt{t}} \leq 2\sqrt{T} - 1$

(2) (10 points) Show the regret of OGD if we choose $\eta_t \propto \frac{1}{\sqrt{t}}$. [Hint: use the above inequality.]

5. (20 points) Recall FTL for number-guessing game. Extend the previous algorithm and analysis to the vector case when the adversary selects a vector $\mathbf{y}_t \in \mathbb{R}^d, \|\mathbf{y}_t\|_2 \leq 1$, the algorithm guesses a vector $\mathbf{x}_t \in \mathbb{R}^d$, and the loss function is $(\mathbf{x}_t - \mathbf{y}_t)^2$. Show an upper bound to the regret logarithmic in T that is independent from d .

6. (20 points) Assuming $\|\mathbf{g}_t\|_2 \leq L$ and $\|\mathbf{x}\|_2 \leq D, \forall \mathbf{x} \in V$. Show that FTRL with regularization $\psi(w) = \frac{1}{2\eta} \|\mathbf{x}\|_2^2$ for linear losses $\ell_t(x) = \mathbf{g}_t^\top \mathbf{x}, \mathbf{x} \in V$ achieves a sublinear regret.